

B6 Condensed-Matter Physics — Model Solutions, Trinity 2023
Paper A13485W1

Question 1 — Neutron diffraction from NaTaO₃

(a) Systematic absences from Na, Ta, O₁

Define the species structure factor:

$$F_X(hkl) = b_X \sum_{j=1}^{N_X} \exp[2\pi i(hx_j + ky_j + lz_j)]. \quad (1)$$

Na at $(0, \frac{1}{2}, \frac{1}{2})$ and $(\frac{1}{2}, 0, \frac{1}{2})$:

$$F_{\text{Na}} = b_{\text{Na}} [e^{2\pi i(k/2+l/2)} + e^{2\pi i(h/2+l/2)}]. \quad (2)$$

Pull out $e^{i\pi l}$:

$$F_{\text{Na}} = b_{\text{Na}} e^{i\pi l} [e^{i\pi k} + e^{i\pi h}]. \quad (3)$$

Ta at $(0, 0, 0)$ and $(\frac{1}{2}, \frac{1}{2}, 0)$:

$$F_{\text{Ta}} = b_{\text{Ta}} [1 + e^{i\pi(h+k)}]. \quad (4)$$

O₁ at $(0, 0, \frac{1}{2})$ and $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$:

$$F_{\text{O1}} = b_{\text{O}} e^{i\pi l} [1 + e^{i\pi(h+k)}]. \quad (5)$$

Take $h + k = 2n + 1$ (odd): $e^{i\pi(h+k)} = -1$:

$$F_{\text{Ta}} = 0, \quad F_{\text{O1}} = 0. \quad (6)$$

Same parity argument: h, k have opposite parity, so $e^{i\pi h} = -e^{i\pi k}$:

$$F_{\text{Na}} = 0. \quad (7)$$

Hence none of Na, Ta, O₁ contributes when $h + k$ is odd. Take $h + k = 2n$: $e^{i\pi h} = e^{i\pi k} = (-1)^h$ and $e^{i\pi(h+k)} = +1$:

$$F_{\text{Na}} = 2b_{\text{Na}}(-1)^h e^{i\pi l} = 2b_{\text{Na}}(-1)^{h+l}, \quad F_{\text{Ta}} = 2b_{\text{Ta}}, \quad F_{\text{O1}} = 2b_{\text{O}}(-1)^l. \quad (8)$$

Sum the three contributions:

$$\boxed{F_{\text{Na+Ta+O1}}(hkl) = 2[b_{\text{Ta}} + b_{\text{O}}(-1)^l + b_{\text{Na}}(-1)^{h+l}] \quad (h + k = 2n).} \quad (9)$$

(b) Structure factor of O₂ to first order in δ

All four O₂ sites have $z = 0$, so $e^{2\pi i l z} = 1$ for every term. Sites:

$$(\frac{1}{4} + \delta, -\frac{1}{4} + \delta, 0), (-\frac{1}{4} - \delta, \frac{1}{4} - \delta, 0), (\frac{1}{4} - \delta, \frac{1}{4} + \delta, 0), (-\frac{1}{4} + \delta, -\frac{1}{4} - \delta, 0). \quad (10)$$

Group sites 1,2 (conjugates) and sites 3,4 (conjugates):

$$F_{O2} = 2b_O \left\{ \cos\left[\frac{\pi(h-k)}{2} + 2\pi\delta(h+k)\right] + \cos\left[\frac{\pi(h+k)}{2} + 2\pi\delta(k-h)\right] \right\}. \quad (11)$$

Linearise with $\cos(\theta + \epsilon) \approx \cos\theta - \epsilon \sin\theta$:

$$\frac{F_{O2}}{2b_O} \approx \cos\frac{\pi(h-k)}{2} + \cos\frac{\pi(h+k)}{2} - 2\pi\delta \left[(h+k) \sin\frac{\pi(h-k)}{2} + (k-h) \sin\frac{\pi(h+k)}{2} \right]. \quad (12)$$

(i) h, k both odd. Then $(h \pm k)/2$ are integers, so both sines vanish. The cosines are $(-1)^{(h-k)/2}$ and $(-1)^{(h+k)/2}$; since $(h+k)/2 - (h-k)/2 = k$ (odd), they have opposite signs:

$$\boxed{F_{O2} = 0 \quad (h, k \text{ both odd}).} \quad (13)$$

(ii) h, k both even. Sines vanish; $(h-k)/2$ and $(h+k)/2$ differ by k (even), so the two cosines are equal:

$$\boxed{F_{O2} = 4b_O (-1)^{(h+k)/2} \quad (h, k \text{ both even}).} \quad (14)$$

(iii) One of h, k even, the other odd. Then $(h \pm k)/2$ are half-integers, so both cosines vanish; only the δ -linear term survives. Write $s_1 = \sin\frac{\pi(h-k)}{2}$ and $s_2 = \sin\frac{\pi(h+k)}{2}$. If k is even (h odd): $s_2 = s_1$, so:

$$F_{O2} = -4\pi\delta b_O [(h+k) + (k-h)] s_1 = -8\pi\delta b_O k s_1. \quad (15)$$

If k is odd (h even): $s_2 = -s_1$, so:

$$F_{O2} = -4\pi\delta b_O [(h+k) - (k-h)] s_1 = -8\pi\delta b_O h s_1. \quad (16)$$

(c) Powder diffraction at $\lambda = 0.159$ nm

Apply $d^{-2} = (h^2 + k^2)/a^2 + l^2/c^2$:

$$d_{001} = c = 0.393 \text{ nm}, \quad (17)$$

$$d_{110} = a/\sqrt{2} = 0.555 \text{ nm}/\sqrt{2} = 0.39245 \text{ nm}, \quad (18)$$

$$d_{210} = a/\sqrt{5} = 0.555 \text{ nm}/\sqrt{5} = 0.24820 \text{ nm}. \quad (19)$$

Apply Bragg $\sin\theta = \lambda/(2d)$:

$$2\theta_{001} = 2 \arcsin(0.159/0.786) = 2 \arcsin(0.20229) = 23.34^\circ, \quad (20)$$

$$2\theta_{110} = 2 \arcsin(0.159/0.7849) = 2 \arcsin(0.20257) = 23.38^\circ, \quad (21)$$

$$2\theta_{210} = 2 \arcsin(0.159/0.4964) = 2 \arcsin(0.32034) = 37.36^\circ. \quad (22)$$

Compare splitting to resolution:

$$|2\theta_{110} - 2\theta_{001}| = 0.04^\circ < 0.05^\circ, \quad (23)$$

so (001) and (110) merge into one peak.

(001) **structure factor.** Here $h + k = 0$ (even), h, k both even. Use (??) with $h = k = 0, l = 1$:

$$F_{\text{Na+Ta+O1}} = 2[b_{\text{Ta}} - b_{\text{O}} - b_{\text{Na}}]. \quad (24)$$

Plug in numbers:

$$F_{\text{Na+Ta+O1}} = 2(6.91 - 5.803 - 3.63) \text{ fm} = -5.046 \text{ fm}. \quad (25)$$

Use (??) with $(h + k)/2 = 0$:

$$F_{\text{O2}} = 4b_{\text{O}} = 4 \cdot 5.803 \text{ fm} = 23.212 \text{ fm}. \quad (26)$$

Sum:

$$F(001) = -5.046 + 23.212 = 18.166 \text{ fm}, \quad |F(001)|^2 \approx 330 \text{ fm}^2. \quad (27)$$

(110) **structure factor.** Here $h + k = 2$ (even), h, k both odd. Use (??) with $h = k = 1, l = 0$:

$$F_{\text{Na+Ta+O1}} = 2[b_{\text{Ta}} + b_{\text{O}} - b_{\text{Na}}]. \quad (28)$$

Plug in numbers:

$$F_{\text{Na+Ta+O1}} = 2(6.91 + 5.803 - 3.63) \text{ fm} = 18.166 \text{ fm}. \quad (29)$$

Apply (??):

$$F_{\text{O2}} = 0, \quad |F(110)|^2 = (18.166)^2 \text{ fm}^2 \approx 330 \text{ fm}^2. \quad (30)$$

(210) **structure factor.** Here $h + k = 3$ (odd), so $F_{\text{Na+Ta+O1}} = 0$. With $h = 2, k = 1$, use (??) with $s_1 = \sin \frac{\pi(1)}{2} = +1$:

$$F(210) = -8\pi\delta b_{\text{O}} \cdot k \cdot s_1 = -8\pi\delta b_{\text{O}}. \quad (31)$$

Square it:

$$|F(210)|^2 = 64\pi^2\delta^2 b_{\text{O}}^2 = 64\pi^2\delta^2 \cdot 33.675 \text{ fm}^2 \approx 2.128 \times 10^4 \delta^2 \text{ fm}^2. \quad (32)$$

Tetragonal multiplicities (4/mmm).

- (00l): only $\pm l$; $m_{001} = 2$.
- (hh0), $h \neq 0$: $(\pm h, \pm h, 0)$; $m_{110} = 4$.
- (hk0), $h \neq k \neq 0$: $(\pm h, \pm k, 0)$ and $(\pm k, \pm h, 0)$; $m_{210} = 8$.

(hkl)	$ F ^2 \text{ (fm}^2\text{)}$	multiplicity	2θ
(001)	330	2	23.34°
(110)	330	4	23.38°
(210)	$2.13 \times 10^4 \delta^2$	8	37.36°

(d) Determination of δ

Write powder intensity as $I_{hkl} = K m_{hkl} |F_{hkl}|^2$. Use $|F(001)|^2 = |F(110)|^2 = 330 \text{ fm}^2$:

$$I_{001} + I_{110} = K(m_{001} + m_{110}) \cdot 330 \text{ fm}^2 = K \cdot 6 \cdot 330 \text{ fm}^2 = 1980 K \text{ fm}^2. \quad (33)$$

Set equal to the measured 1843:

$$K = \frac{1843}{1980} \text{ fm}^{-2} \approx 0.9308 \text{ fm}^{-2}. \quad (34)$$

Use $I_{210} = K m_{210} |F(210)|^2$ with $m_{210} = 8$:

$$I_{210} = K \cdot 8 \cdot 64\pi^2 \delta^2 b_O^2 = 512\pi^2 K b_O^2 \delta^2. \quad (35)$$

Solve for δ^2 with $b_O^2 = 33.675 \text{ fm}^2$, $K = 0.9308$, $I_{210} = 345$:

$$\delta^2 = \frac{345}{512\pi^2 \cdot 0.9308 \cdot 33.675}. \quad (36)$$

Evaluate denominator:

$$512\pi^2 \cdot 0.9308 \cdot 33.675 = 1.583 \times 10^5. \quad (37)$$

Hence:

$$\delta^2 = \frac{345}{1.583 \times 10^5} = 2.18 \times 10^{-3}. \quad (38)$$

Take the square root:

$$\boxed{\delta \approx 0.047.} \quad (39)$$

Question 2 — Molecular orbital theory of H_2

(a) Secular equation retaining overlap

Write the variational quotient:

$$\lambda = \frac{N}{D}, \quad N = \sum_{ij} c_i^* c_j H_{ij}, \quad D = \sum_{ij} c_i^* c_j S_{ij}, \quad (40)$$

with $H_{ij} = \langle i | \hat{H} | j \rangle$ and $S_{ij} = \langle i | j \rangle$. Apply $(\hat{K} + \hat{V}_1) |1\rangle = \epsilon_0 |1\rangle$:

$$\hat{H} |1\rangle = (\hat{K} + \hat{V}_1) |1\rangle + \hat{V}_2 |1\rangle = \epsilon_0 |1\rangle + \hat{V}_2 |1\rangle. \quad (41)$$

Take $\langle 1 |$:

$$H_{11} = \epsilon_0 + \langle 1 | \hat{V}_2 | 1 \rangle = \epsilon_0 + V_{c2}. \quad (42)$$

By analogy ($1 \leftrightarrow 2$):

$$H_{22} = \epsilon_0 + V_{c1}. \quad (43)$$

Apply $(\hat{K} + \hat{V}_2) |2\rangle = \epsilon_0 |2\rangle$:

$$\hat{H} |2\rangle = \epsilon_0 |2\rangle + \hat{V}_1 |2\rangle. \quad (44)$$

Take $\langle 1 |$:

$$H_{12} = \epsilon_0 \langle 1 | 2 \rangle + \langle 1 | \hat{V}_1 | 2 \rangle = \epsilon_0 \sigma - t, \quad (45)$$

using $t = -\langle 1 | \hat{V}_2 | 2 \rangle$ and the convention $\langle 1 | \hat{V}_1 | 2 \rangle = -t$ that follows from the symmetric problem (see (b)). By Hermiticity:

$$H_{21} = \epsilon_0 \sigma^* - t^*. \quad (46)$$

Assemble:

$$\mathbf{H} = \begin{pmatrix} \epsilon_0 + V_{c2} & \epsilon_0\sigma - t \\ \epsilon_0\sigma^* - t^* & \epsilon_0 + V_{c1} \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} 1 & \sigma \\ \sigma^* & 1 \end{pmatrix}. \quad (47)$$

Extremise $\lambda = N/D$ via $\partial\lambda/\partial c_i^* = 0$:

$$\frac{\partial N}{\partial c_i^*} = \lambda \frac{\partial D}{\partial c_i^*} \iff \sum_j H_{ij} c_j = \lambda \sum_j S_{ij} c_j. \quad (48)$$

Cast into matrix form:

$$\boxed{(\mathbf{H} - \lambda \mathbf{S}) \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = 0.} \quad (49)$$

(b) Symmetric two-atom case

Apply $x \rightarrow -x$ which swaps $1 \leftrightarrow 2$:

$$V_{c1} = \langle 2 | \hat{V}_1 | 2 \rangle \xrightarrow{1 \leftrightarrow 2} \langle 1 | \hat{V}_2 | 1 \rangle = V_{c2}, \quad (50)$$

hence $V_{c1} = V_{c2} \equiv V_c$. Choose real 1s orbitals; the operators $\hat{V}_i$ are real, so $\sigma, t \in \mathbb{R}$. Insert into (??):

$$\begin{pmatrix} \epsilon_0 + V_c - \lambda & (\epsilon_0 - \lambda)\sigma - t \\ (\epsilon_0 - \lambda)\sigma - t & \epsilon_0 + V_c - \lambda \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = 0. \quad (51)$$

Diagonal entries equal and off-diagonal entries equal, so eigenvectors are $c_1 = \pm c_2$. Substitute $c_1 = +c_2$:

$$(\epsilon_0 + V_c - \lambda) + (\epsilon_0 - \lambda)\sigma - t = 0. \quad (52)$$

Collect λ :

$$\lambda(1 + \sigma) = \epsilon_0(1 + \sigma) + V_c - t. \quad (53)$$

Solve for λ :

$$\boxed{\lambda_+ = \epsilon_0 + \frac{V_c - t}{1 + \sigma} \text{ (bonding).}} \quad (54)$$

Substitute $c_1 = -c_2$:

$$(\epsilon_0 + V_c - \lambda) - [(\epsilon_0 - \lambda)\sigma - t] = 0. \quad (55)$$

Collect λ :

$$\lambda(1 - \sigma) = \epsilon_0(1 - \sigma) + V_c + t. \quad (56)$$

Solve for λ :

$$\boxed{\lambda_- = \epsilon_0 + \frac{V_c + t}{1 - \sigma} \text{ (antibonding).}} \quad (57)$$

Normalise: $\langle \psi_{\pm} | \psi_{\pm} \rangle = 2(1 \pm \sigma)$ (using $\langle 1 | 2 \rangle = \sigma$):

$$|\psi_+\rangle = \frac{|1\rangle + |2\rangle}{\sqrt{2(1 + \sigma)}}, \quad |\psi_-\rangle = \frac{|1\rangle - |2\rangle}{\sqrt{2(1 - \sigma)}}. \quad (58)$$

(c) Hydrogen molecule with explicit integrals

(i) Large- d asymptotics. Write t, σ as bilinear overlaps of $\phi(r) = \pi^{-1/2}e^{-r}$: the product $\phi(\mathbf{r})\phi(\mathbf{r}-\mathbf{d}) \sim e^{-r}e^{-|\mathbf{r}-\mathbf{d}|}$ is exponentially suppressed for any fixed point unless $\mathbf{r}$ sits in a narrow region between the two centres, so:

$$t, \sigma \propto e^{-d} \quad (d \gg a_0). \quad (59)$$

By contrast $V_c = \langle 1 | \hat{V}_2 | 1 \rangle$ is the electrostatic energy of $|\phi_1|^2$ (total charge 1, centred at site 1) in the Coulomb potential of the distant proton at site 2. For $d \gg a_0$ the charge $|\phi_1|^2$ looks like a point charge:

$$V_c \rightarrow -2/d \quad (d \gg a_0). \quad (60)$$

(ii) Ground-state energy at $d_{\text{exp}} = 1.4$. Compute t :

$$t = 2e^{-1.4}(1 + 1.4) = 2 \cdot 0.2466 \cdot 2.4 = 1.184 E_R. \quad (61)$$

Compute σ :

$$\sigma = e^{-1.4}(1 + 1.4 + 1.96/3) = 0.2466 \cdot 3.0533 = 0.7530. \quad (62)$$

Compute V_c :

$$V_c = \frac{2[e^{-2.8}(1 + 1.4) - 1]}{1.4} = \frac{2(0.06081 \cdot 2.4 - 1)}{1.4} = \frac{2(-0.8541)}{1.4} = -1.220 E_R. \quad (63)$$

Substitute into (??) with $\epsilon_0 = -1$:

$$\lambda_+^{(\sigma)} = -1 + \frac{-1.220 - 1.184}{1 + 0.7530} = -1 + \frac{-2.404}{1.753} = -1 - 1.371 = -2.371 E_R. \quad (64)$$

Set $\sigma \rightarrow 0$:

$$\lambda_+^{(\sigma=0)} = -1 + (-1.220 - 1.184) = -3.404 E_R. \quad (65)$$

Place two electrons in $|\psi_+\rangle$ with opposite spin and add proton repulsion $2/d$ (in E_R):

$$E_{\text{tot}}^{(\sigma)} = 2\lambda_+^{(\sigma)} + 2/d = 2(-2.371) + 2/1.4 = -4.742 + 1.4286 = -3.314 E_R, \quad (66)$$

$$E_{\text{tot}}^{(\sigma=0)} = 2(-3.404) + 1.4286 = -5.379 E_R. \quad (67)$$

Subtract the dissociation limit $E_{2\text{H}} = 2\epsilon_0 = -2 E_R$:

$$E_d^{(\sigma)} = -2 - (-3.314) = 1.31 E_R, \quad E_d^{(\sigma=0)} = -2 - (-5.379) = 3.38 E_R. \quad (68)$$

The two values differ by a factor of ~ 2.6 , so σ cannot be neglected.

(iii) Comparison with experiment. The MO prediction $E_d^{(\sigma)} = 1.31 E_R$ greatly exceeds $E_d^{\text{exp}} \approx 0.33 E_R$: the MO calculation overbinds by $\sim 1 E_R \approx 13.6 \text{ eV}$.

(iv) **Behaviour at $d = 0$.** Take $d \rightarrow 0$ in t :

$$t(0) = 2 \cdot 1 \cdot (1 + 0) = 2 E_R. \quad (69)$$

Take $d \rightarrow 0$ in σ :

$$\sigma(0) = 1 \cdot (1 + 0 + 0) = 1. \quad (70)$$

Take $d \rightarrow 0$ in V_c : expand $e^{-2d}(1 + d) = (1 - 2d + 2d^2)(1 + d) = 1 - d + O(d^3)$:

$$V_c(0) = \lim_{d \rightarrow 0} \frac{2(-d + O(d^3))}{d} = -2 E_R. \quad (71)$$

Insert into (??):

$$\lambda_+(0) = -1 + \frac{-2 - 2}{1 + 1} = -1 - 2 = -3 E_R. \quad (72)$$

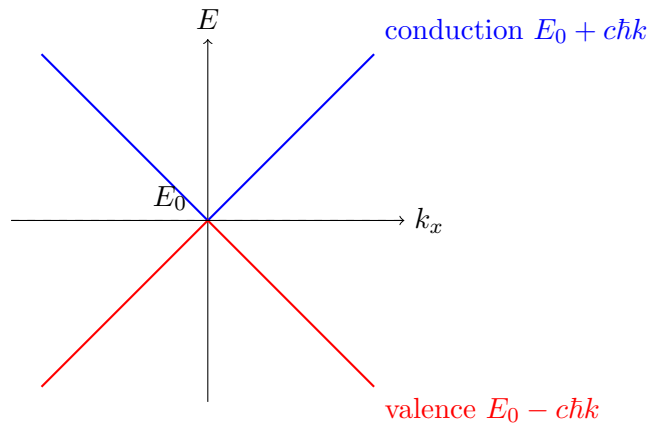
Hence $\lambda_+(0) = -3 E_R < \lambda_+(d_{\text{exp}}) = -2.371 E_R$: λ_+ keeps decreasing as $d \rightarrow 0$, with absolute minimum at $d = 0$.

(d) Missing physics

The MO calculation overestimates the dissociation energy by a factor ~ 4 at d_{exp} and predicts an unphysical minimum of λ_+ at $d = 0$. Both flaws stem from omitted *electron–electron repulsion*. The trial state places two electrons in the same spatial orbital $|\psi_+\rangle$; squeezing the molecule to small d forces both electrons into the same small region and the Coulomb energy $\sim 1/r_{12}$ grows. Adding the Hartree term $J = \langle \psi_+ \psi_+ | 1/r_{12} | \psi_+ \psi_+ \rangle > 0$ raises the energy strongly at small d (removing the spurious $d = 0$ minimum in the total energy) and weakens the binding at d_{exp} . A related deficiency is the absence of *electron correlation*: the closed-shell MO state retains equal amplitudes for ionic (H^-H^+) and covalent configurations as $d \rightarrow \infty$, so dissociation is described poorly. Heitler–London or CI with the antibonding orbital cures this and pushes the binding curve toward the experimental $0.33 E_R$.

Question 3 — Massless Dirac fermions in 2D

(a) **Energy surfaces and comparison with $E_g \rightarrow 0$ semiconductor**



(ii) Comparison with a parabolic $E_g \rightarrow 0$ semiconductor.

- Dispersion: linear (here) vs. quadratic. Group velocity here is energy-independent: $|\mathbf{v}_g| = c$.
- Density of states: $g(E) \propto |E - E_0|$ (vanishes at Dirac point) vs. constant in 2D parabolic.
- Effective mass: undefined for linear bands; finite for parabolic.
- Berry phase: π around the Dirac point (suppresses backscattering); absent in parabolic case.

(b) Conduction-band electron density

Count states (factor 2 for spin) in a k -shell of area $2\pi k dk$:

$$g(E) dE = 2 \cdot \frac{2\pi k dk}{(2\pi)^2} = \frac{k}{\pi} dk. \quad (73)$$

Substitute $k = (E - E_0)/(c\hbar)$, $dk = dE/(c\hbar)$:

$$g(E) = \frac{E - E_0}{\pi c^2 \hbar^2}. \quad (74)$$

Apply the Fermi function:

$$n_{\text{el}} = \int_{E_0}^{\infty} \frac{(E - E_0) dE}{\pi c^2 \hbar^2 [e^{\beta(E-\mu)} + 1]}. \quad (75)$$

Change variables $u = \beta(E - E_0)$, $du = \beta dE$:

$$n_{\text{el}} = \frac{1}{\pi c^2 \hbar^2 \beta^2} \int_0^{\infty} \frac{u du}{e^{u-\alpha} + 1}. \quad (76)$$

Recognise the Fermi integral with $j = 1$:

$$\int_0^{\infty} \frac{u du}{e^{u-\alpha} + 1} = 1! F_1(\alpha) = F_1(\alpha). \quad (77)$$

Hence:

$$\boxed{n_{\text{el}} = \frac{1}{\pi c^2 \hbar^2} \beta^{-2} F_1(\alpha), \quad C_1 = \frac{1}{\pi c^2 \hbar^2}, \quad p = -2.} \quad (78)$$

(c) Conduction-band specific heat

Energy density above E_0 :

$$\tilde{u}_{\text{el}} = \int_{E_0}^{\infty} (E - E_0) g(E) f(E) dE. \quad (79)$$

Substitute $u = \beta(E - E_0)$ again:

$$\tilde{u}_{\text{el}} = \frac{1}{\pi c^2 \hbar^2 \beta^3} \int_0^{\infty} \frac{u^2 du}{e^{u-\alpha} + 1}. \quad (80)$$

Use $\int_0^{\infty} u^2 du / (e^{u-\alpha} + 1) = 2! F_2(\alpha) = 2F_2(\alpha)$:

$$\tilde{u}_{\text{el}} = \frac{2}{\pi c^2 \hbar^2 \beta^3} F_2(\alpha) = \frac{2(k_B T)^3}{\pi c^2 \hbar^2} F_2(\alpha). \quad (81)$$

At fixed μ (so fixed $\tilde{\mu} = \mu - E_0$):

$$\left. \frac{d\alpha}{dT} \right|_{\mu} = \frac{d}{dT} \left(\frac{\tilde{\mu}}{k_B T} \right) = -\frac{\tilde{\mu}}{k_B T^2} = -\frac{\alpha}{T}. \quad (82)$$

Differentiate (??) using $dF_2/d\alpha = F_1$:

$$c_v^{\text{el}} = \left. \frac{d\tilde{u}_{\text{el}}}{dT} \right|_{\mu} = \frac{2}{\pi c^2 \hbar^2} \left[3k_B^3 T^2 F_2(\alpha) + (k_B T)^3 F_1(\alpha) \frac{d\alpha}{dT} \right]. \quad (83)$$

Insert (??):

$$c_v^{\text{el}} = \frac{2}{\pi c^2 \hbar^2} \left[3k_B^3 T^2 F_2(\alpha) - k_B^3 T^2 \alpha F_1(\alpha) \right]. \quad (84)$$

Factor $k_B^3 T^2$:

$$\boxed{c_v^{\text{el}} = \frac{2}{\pi c^2 \hbar^2} k_B^3 T^2 [3F_2(\alpha) - \alpha F_1(\alpha)], \quad C_2 = \frac{2}{\pi c^2 \hbar^2}, \quad q = 2, \quad a = 3, \quad b = -1.} \quad (85)$$

(d) Charge neutrality at $\tilde{\mu} = 0$ and total specific heat

(i) Equal electron and hole numbers. The valence band has $g_v(E) = (E_0 - E)/(\pi c^2 \hbar^2)$ for $E \leq E_0$ (reflection of (??) about E_0). Hole occupation:

$$1 - f(E) = \frac{1}{e^{\beta(\mu - E)} + 1}. \quad (86)$$

Hole density:

$$n_h = \int_{-\infty}^{E_0} g_v(E) [1 - f(E)] dE. \quad (87)$$

Substitute $u = \beta(E_0 - E)$:

$$n_h = \frac{1}{\pi c^2 \hbar^2 \beta^2} \int_0^{\infty} \frac{u du}{e^{u+\alpha} + 1} = \frac{F_1(-\alpha)}{\pi c^2 \hbar^2 \beta^2}. \quad (88)$$

Set $\alpha = 0$:

$$n_h = \frac{F_1(0)}{\pi c^2 \hbar^2 \beta^2} = n_{\text{el}}. \quad (89)$$

By the same $\alpha \rightarrow -\alpha$ symmetry, the hole specific heat equals the electron specific heat at $\alpha = 0$.

(ii) Total specific heat at $\tilde{\mu} = 0$. Sum and set $\alpha = 0$ (so $\alpha F_1 = 0$):

$$c_v^{\text{tot}}|_{\tilde{\mu}=0} = 2 \cdot \frac{2k_B^3 T^2}{\pi c^2 \hbar^2} \cdot 3F_2(0) = \frac{12k_B^3 T^2 F_2(0)}{\pi c^2 \hbar^2}. \quad (90)$$

Read $F_2(0) = 3\zeta/4$ from the small- α expansion:

$$\boxed{c_v^{\text{tot}}|_{\tilde{\mu}=0} = \frac{12 \cdot 3\zeta/4}{\pi c^2 \hbar^2} k_B^3 T^2 = \frac{9\zeta k_B^3 T^2}{\pi c^2 \hbar^2}.} \quad (91)$$

Plug in $k_B = 1.381 \times 10^{-23} \text{ J/K}$, $\hbar = 1.055 \times 10^{-34} \text{ J s}$, $c = 10^6 \text{ m/s}$, $T = 100 \text{ K}$:

$$\pi c^2 \hbar^2 = \pi \cdot 10^{12} \cdot (1.055 \times 10^{-34})^2 = 3.497 \times 10^{-56} \text{ J}^2 \text{ m}^2. \quad (92)$$

$$k_B^3 T^2 = (1.381 \times 10^{-23})^3 \cdot 10^4 = 2.634 \times 10^{-65} \text{ J}^3/\text{K}. \quad (93)$$

$$c_v^{\text{tot}} = \frac{9 \cdot 1.2021 \cdot 2.634 \times 10^{-65}}{3.497 \times 10^{-56}} \approx 8.15 \times 10^{-9} \text{ J K}^{-1} \text{ m}^{-2}. \quad (94)$$

(e) **Degenerate limit** $\tilde{\mu} \gg k_B T$

Differentiate the given $F_2(\alpha) \approx \alpha^3/6 + \pi^2\alpha/6$ to get F_1 :

$$F_1(\alpha) = \frac{dF_2}{d\alpha} \approx \frac{\alpha^2}{2} + \frac{\pi^2}{6}. \quad (95)$$

(i) **Temperature dependence of n_{el} .** Insert into (??):

$$n_{\text{el}} = \frac{(k_B T)^2}{\pi c^2 \hbar^2} \left(\frac{\alpha^2}{2} + \frac{\pi^2}{6} \right). \quad (96)$$

Use $\alpha k_B T = \tilde{\mu}$:

$$n_{\text{el}} = \frac{1}{\pi c^2 \hbar^2} \left[\frac{\tilde{\mu}^2}{2} + \frac{\pi^2}{6} (k_B T)^2 \right]. \quad (97)$$

(ii) **Chemical potential at $T \rightarrow 0$.** Drop the T^2 term:

$$n_{\text{el}} = \frac{\tilde{\mu}^2}{2\pi c^2 \hbar^2}. \quad (98)$$

Solve for $\tilde{\mu}$:

$$\tilde{\mu} = \hbar c \sqrt{2\pi n_{\text{el}}}. \quad (99)$$

Restore $\mu = \tilde{\mu} + E_0$:

$$\mu \rightarrow E_0 + \hbar c \sqrt{2\pi n_{\text{el}}}^{1/2}. \quad (100)$$

Read off:

$$C_3 = \hbar c \sqrt{2\pi}, \quad r = \frac{1}{2}. \quad (101)$$

Question 4 — Easy-axis anisotropy with transverse field, $j = 1$

(a) **Matrix representation in the $j = 1$ basis**

Work in $\{|1, 1\rangle, |1, 0\rangle, |1, -1\rangle\}$. List $\hat{J}^2$:

$$\hat{J}^2 = j(j+1)\hbar^2 \mathbb{K} = 2\hbar^2 \mathbb{K}, \quad (102)$$

setting $\hbar = 1$ for the numerical coefficients. List $\hat{J}_z$ and its square:

$$\hat{J}_z = \text{diag}(1, 0, -1), \quad \hat{J}_z^2 = \text{diag}(1, 0, 1). \quad (103)$$

Compute $\hat{J}^2 - \hat{J}_z^2$:

$$\hat{J}^2 - \hat{J}_z^2 = \text{diag}(2-1, 2-0, 2-1) = \text{diag}(1, 2, 1). \quad (104)$$

Halve it:

$$\frac{1}{2}(\hat{J}^2 - \hat{J}_z^2) = \text{diag}(\frac{1}{2}, 1, \frac{1}{2}). \quad (105)$$

Compute $\hat{J}_+^2 |1, -1\rangle$ using $\hat{J}_+ |1, m\rangle = \sqrt{(1-m)(2+m)} |1, m+1\rangle$:

$$\hat{J}_+ |1, -1\rangle = \sqrt{2} |1, 0\rangle, \quad \hat{J}_+^2 |1, -1\rangle = \sqrt{2} \hat{J}_+ |1, 0\rangle = 2 |1, 1\rangle. \quad (106)$$

All other actions of $\hat{J}_+^2$ vanish. By Hermitian conjugation $\hat{J}_-^2 |1, 1\rangle = 2 |1, -1\rangle$, so:

$$\frac{1}{4}(\hat{J}_+^2 + \hat{J}_-^2) = \begin{pmatrix} 0 & 0 & 1/2 \\ 0 & 0 & 0 \\ 1/2 & 0 & 0 \end{pmatrix}. \quad (107)$$

Add the two pieces:

$$\hat{J}_x^2 = \begin{pmatrix} 1/2 & 0 & 1/2 \\ 0 & 1 & 0 \\ 1/2 & 0 & 1/2 \end{pmatrix}. \quad (108)$$

Match to the given form:

$$-D\hat{J}_x^2 = -D \begin{pmatrix} a & 0 & a \\ 0 & 2a & 0 \\ a & 0 & a \end{pmatrix} \Rightarrow \boxed{a = \frac{1}{2}}. \quad (109)$$

Match the Zeeman term:

$$-g\mu_B B \hat{J}_z = -g\mu_B B \text{diag}(b, 0, -b) = -g\mu_B B \text{diag}(1, 0, -1) \Rightarrow \boxed{b = 1}. \quad (110)$$

(b) Eigenvalues to lowest order in B

The middle row/column decouples; eigenvalue is

$$E_2 = -D \cdot 2a = -D. \quad (111)$$

The remaining 2×2 block in $\{|1, 1\rangle, |1, -1\rangle\}$:

$$\mathbf{H}' = \begin{pmatrix} -D/2 - g\mu_B B & -D/2 \\ -D/2 & -D/2 + g\mu_B B \end{pmatrix}. \quad (112)$$

Compute the trace:

$$\text{tr } \mathbf{H}' = -D. \quad (113)$$

Compute the determinant:

$$\det \mathbf{H}' = (D/2)^2 - (g\mu_B B)^2 - (D/2)^2 = -(g\mu_B B)^2. \quad (114)$$

Solve the characteristic equation:

$$E_{\pm} = \frac{\text{tr} \pm \sqrt{\text{tr}^2 - 4 \det}}{2} = -\frac{D}{2} \pm \frac{1}{2} \sqrt{D^2 + 4(g\mu_B B)^2}. \quad (115)$$

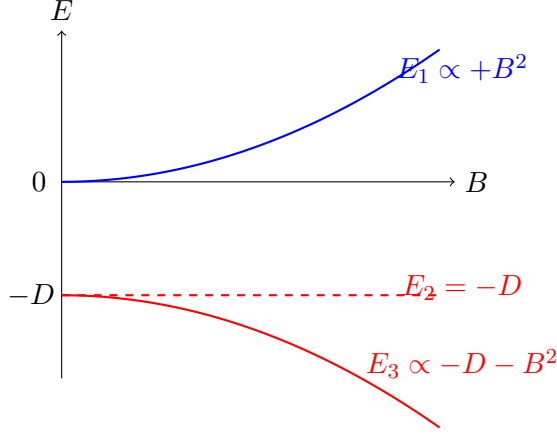
Expand the square root for $g\mu_B B \ll D$:

$$\sqrt{D^2 + 4(g\mu_B B)^2} = D \sqrt{1 + 4(g\mu_B B)^2/D^2} \approx D + \frac{2(g\mu_B B)^2}{D}. \quad (116)$$

Substitute into (??):

$$\boxed{E_1 = +\frac{(g\mu_B B)^2}{D}, \quad E_2 = -D, \quad E_3 = -D - \frac{(g\mu_B B)^2}{D}}. \quad (117)$$

Sketch.



(c) Partition function and transverse susceptibility

(i) Partition function. For N independent ions:

$$Z_N(B) = [Z_1(B)]^N, \quad Z_1 = e^{-\beta E_1} + e^{-\beta E_2} + e^{-\beta E_3}. \quad (118)$$

Introduce $w \equiv (g\mu_B)^2/D$ and insert (??):

$$Z_1(B) = e^{-\beta w B^2} + e^{\beta D} + e^{\beta D + \beta w B^2}. \quad (119)$$

(ii) Magnetic moment in the low-field limit. Expand each exponential to $O(B^2)$:

$$e^{-\beta w B^2} \approx 1 - \beta w B^2, \quad e^{\beta D + \beta w B^2} \approx e^{\beta D} (1 + \beta w B^2). \quad (120)$$

Add the three terms:

$$Z_1(B) \approx 1 + 2e^{\beta D} + \beta w B^2 (e^{\beta D} - 1). \quad (121)$$

Take $\ln$ to leading order in B^2 :

$$\ln Z_1 \approx \ln(1 + 2e^{\beta D}) + \frac{\beta w B^2 (e^{\beta D} - 1)}{1 + 2e^{\beta D}}. \quad (122)$$

Compute the moment per ion:

$$m_1 = -\frac{\partial F_1}{\partial B} = \frac{1}{\beta} \frac{\partial \ln Z_1}{\partial B} = \frac{2wB(e^{\beta D} - 1)}{1 + 2e^{\beta D}} + O(B^3). \quad (123)$$

Multiply by N for the ensemble: $m = Nm_1$.

(iii) Transverse susceptibility. Differentiate at $B \rightarrow 0$:

$$\chi_{\perp} = \mu_0 \left. \frac{dm}{dB} \right|_{B \rightarrow 0} = \frac{2N\mu_0(g\mu_B)^2}{D} \cdot \frac{e^{\beta D} - 1}{1 + 2e^{\beta D}}. \quad (124)$$

Multiply and divide by $\beta D/3$ to factor $\chi_C = 2N\mu_0(g\mu_B)^2\beta/3$:

$$\chi_{\perp} = \chi_C \cdot \frac{3}{D\beta} \cdot \frac{e^{D\beta} - 1}{1 + 2e^{D\beta}}. \quad (125)$$

Read off $f_{\perp}$:

$$\boxed{\chi_{\perp} = \chi_C f_{\perp}(D\beta), \quad f_{\perp}(x) = \frac{3}{x} \cdot \frac{e^x - 1}{1 + 2e^x}.} \quad (126)$$

(d) Limits and comparison with parallel case

(i) Asymptotic $T \rightarrow \infty$ ($x \rightarrow 0$). Expand $e^x - 1 \approx x$, $1 + 2e^x \approx 3$:

$$f_{\parallel}(0) = \frac{3 \cdot 1}{1 + 2} = 1. \quad (127)$$

$$f_{\perp}(0) = \frac{3}{x} \cdot \frac{x}{3} = 1. \quad (128)$$

Hence $\chi_{\parallel}, \chi_{\perp} \rightarrow \chi_C$.

(ii) $\chi_{\parallel} > \chi_C$ at all T . Require $f_{\parallel}(x) > 1$:

$$\frac{3e^x}{1 + 2e^x} > 1 \iff 3e^x > 1 + 2e^x \iff e^x > 1, \quad (129)$$

which holds for all $x > 0$.

(iii) $\chi_{\perp}$ finite at $T = 0$. Take $x \rightarrow \infty$: $e^x - 1 \sim e^x$, $1 + 2e^x \sim 2e^x$:

$$f_{\perp}(x) \rightarrow \frac{3}{x} \cdot \frac{e^x}{2e^x} = \frac{3}{2x}. \quad (130)$$

Multiply by $\chi_C = 2N\mu_0(g\mu_B)^2\beta/3$ and use $\beta = x/D$:

$$\chi_{\perp} \rightarrow \frac{2N\mu_0(g\mu_B)^2\beta}{3} \cdot \frac{3}{2D\beta} = \frac{N\mu_0(g\mu_B)^2}{D}. \quad (131)$$

Finite (Van Vleck).

(e) Interpretation of the figure and determination of D

Physical picture. At high T , $k_B T \gg D$, so all three curves coincide with the Curie tail. At low T , $\chi_{\parallel} \rightarrow (3/2)\chi_C \rightarrow \infty$ while $\chi_{\perp}$ saturates at the temperature-independent Van Vleck value $N\mu_0(g\mu_B)^2/D$. The ground states $|m_x = \pm 1\rangle$ have $\langle \hat{J}_z \rangle = 0$; the transverse field admixes them with $|m_x = 0\rangle$ at energy D above, giving a B -linear induced moment with T -independent coefficient.

Extracting D . Read off the saturation value:

$$\chi_{\perp}^{\text{sat}} \approx 3 \times 10^{-6} \text{ m}^3/\text{mol}. \quad (132)$$

Apply the Van Vleck expression (with $N \rightarrow N_A$ for molar susceptibility):

$$D = \frac{N_A\mu_0(g\mu_B)^2}{\chi_{\perp}^{\text{sat}}}. \quad (133)$$

Compute $(g\mu_B)^2$ with $g = 2$, $\mu_B = 9.274 \times 10^{-24} \text{ J/T}$:

$$(g\mu_B)^2 = (1.8548 \times 10^{-23})^2 = 3.440 \times 10^{-46} \text{ J}^2/\text{T}^2. \quad (134)$$

Compute $N_A\mu_0(g\mu_B)^2$:

$$N_A\mu_0(g\mu_B)^2 = 6.022 \times 10^{23} \cdot 1.2566 \times 10^{-6} \cdot 3.440 \times 10^{-46} = 2.603 \times 10^{-28} \text{ J m}^3/\text{mol}. \quad (135)$$

Divide by $\chi_{\perp}^{\text{sat}}$:

$$D = \frac{2.603 \times 10^{-28}}{3 \times 10^{-6}} \text{ J} = 8.7 \times 10^{-23} \text{ J}. \quad (136)$$

Convert to a temperature:

$$D/k_B = \frac{8.7 \times 10^{-23}}{1.381 \times 10^{-23}} \text{ K} \approx 6.3 \text{ K}. \quad (137)$$

Hence:

$$\boxed{D \approx 8.7 \times 10^{-23} \text{ J} \quad (D/k_B \approx 6 \text{ K}).} \quad (138)$$

Consistency: the curves visibly merge for $T \gtrsim 4D/k_B \approx 25 \text{ K}$, matching $f_{\perp} \approx f_{\parallel} \approx 1$ from part (d)(i).